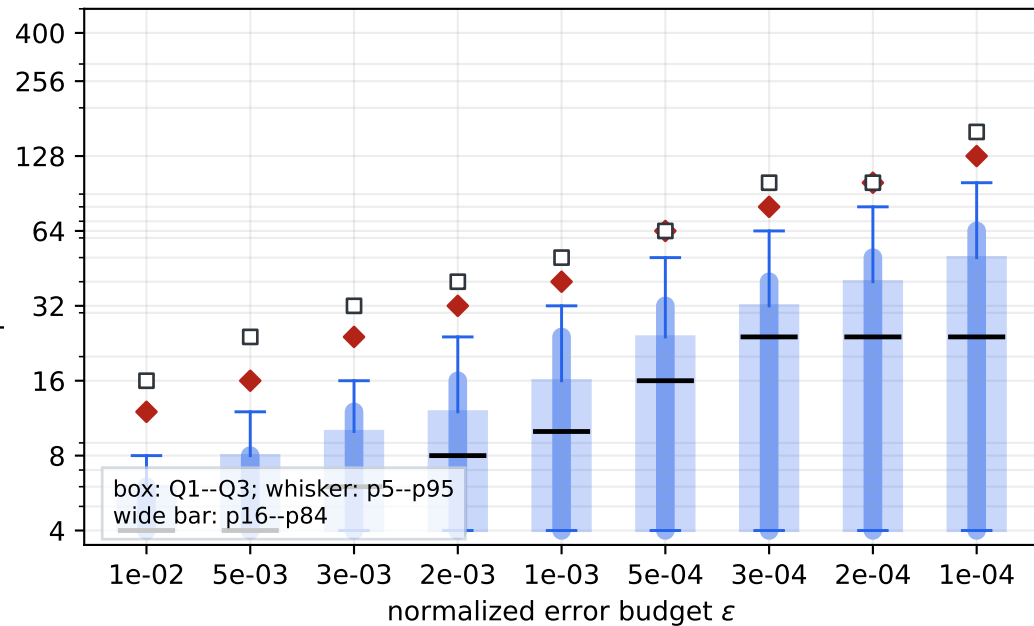
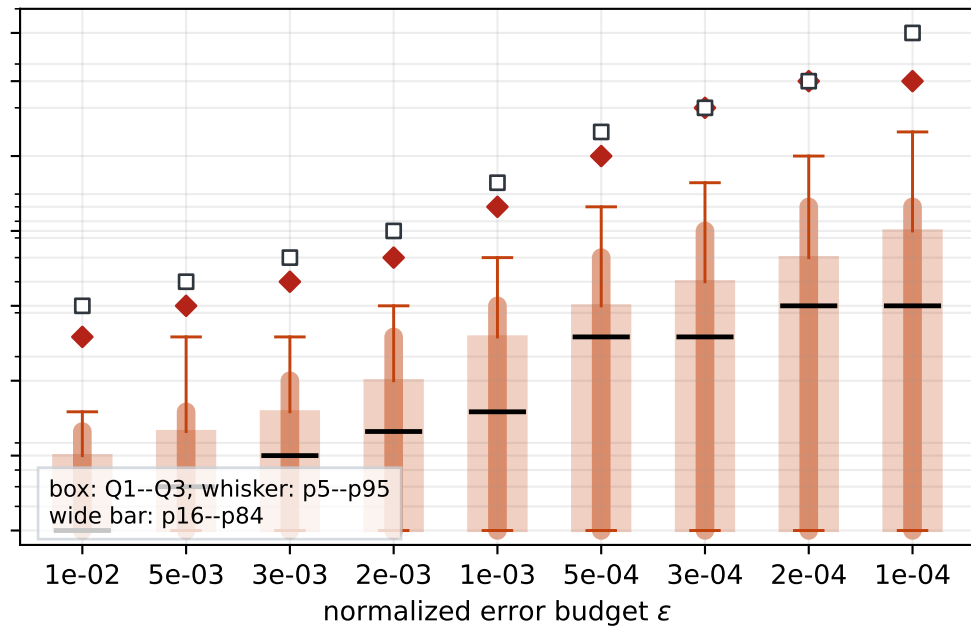


Required resolution: box-and-whisker distribution

cartesian



polar



central 68% (p16--p84)

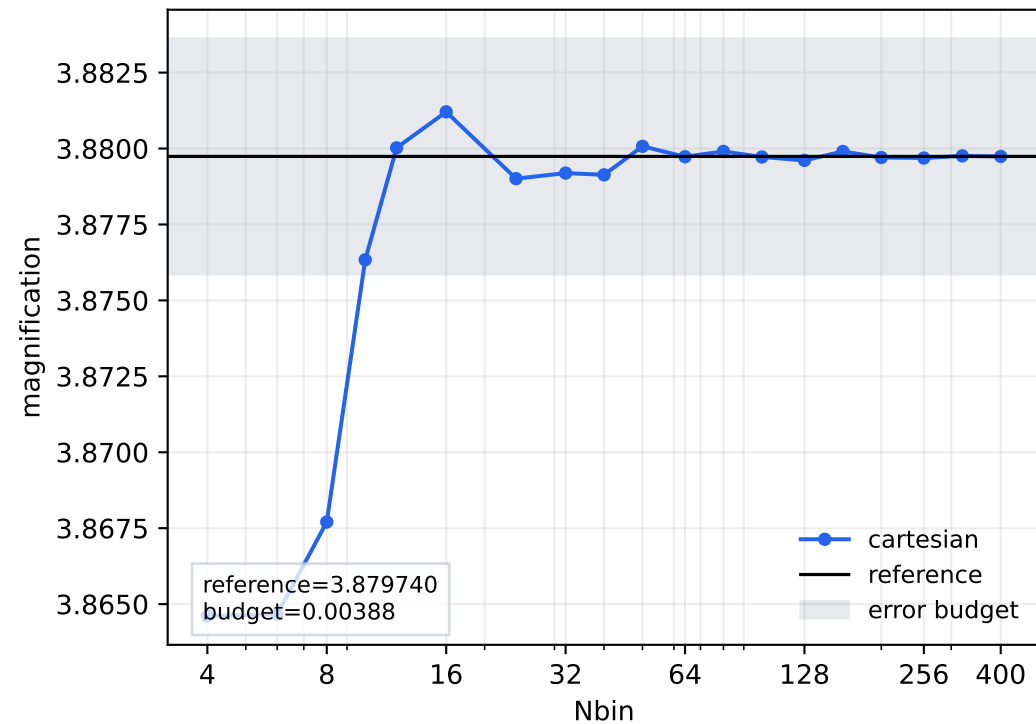
median

p99

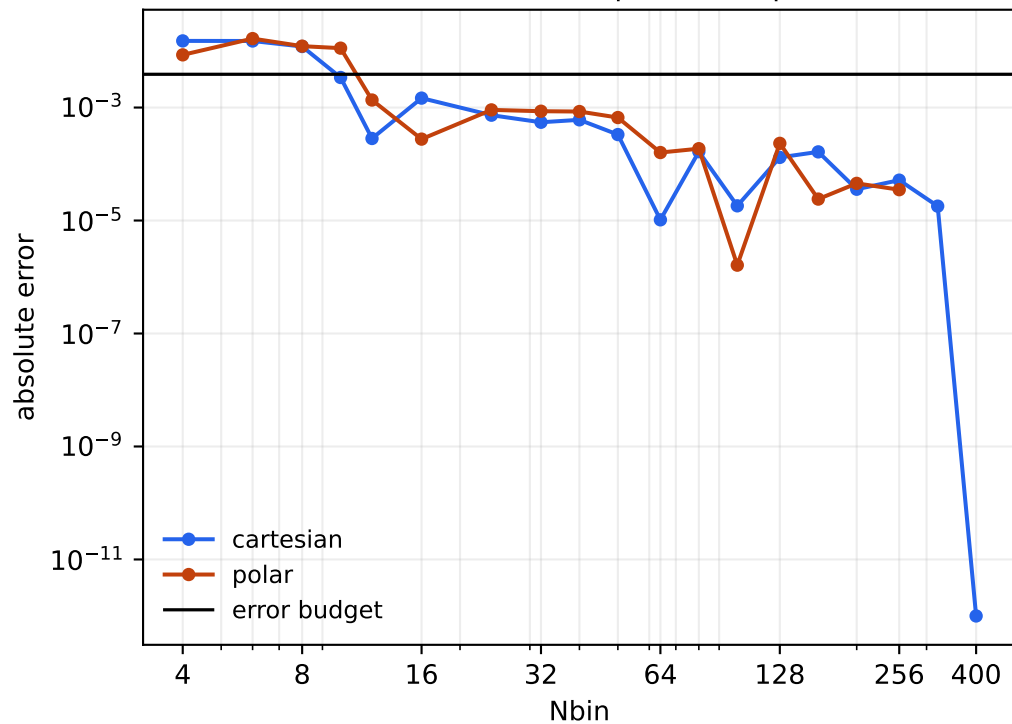
fitted bucket

Representative holdout case: case 48 (linear)
reltol=1e-3; required buckets: Cartesian 10, polar 12

cartesian: A(N)



absolute error $|A(N)-A_{ref}|$



How the empirical law was constructed

- 1. For every case, increase Nbin and record the first persistent crossing of the common error budget against the high-resolution reference.
- 2. At each normalized budget, summarize the required-Nbin distribution: the box is Q1--Q3, whiskers are p5--p95, the wide bar is p16--p84, and p99 is the conservative tail used for calibration.
- 3. Fit the discovery p99 values in log2 space:
 $\log_2(N_{99}) = \log_2(C) + \text{beta} \log_2(1e-3 / \text{epsilon})$.
- 4. Apply the continuous law to the independent holdout, then round upward to the next measured resolution bucket.

grid	C	beta	holdout coverage
cartesian	45.3	0.477	99.67%
polar	94.6	0.595	99.81%

The fit is a population-level p99 rule, not a claim that every case needs N99.